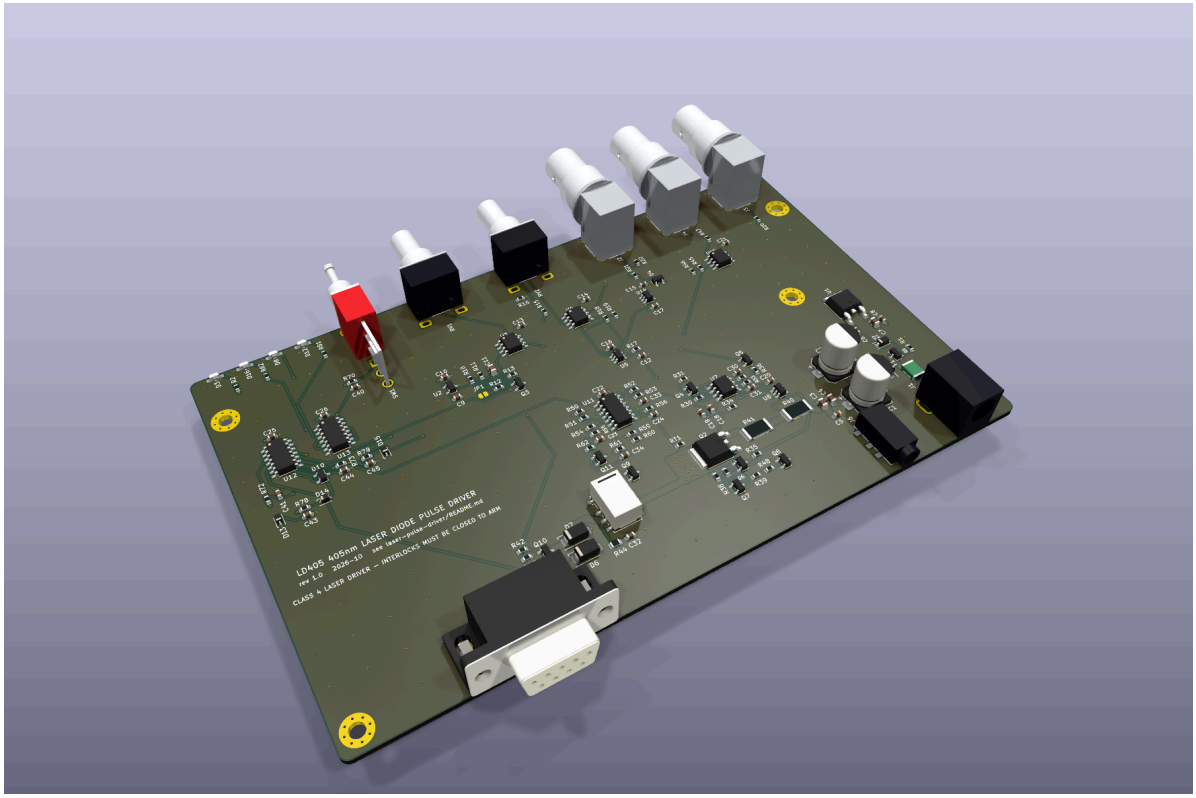
DESIGN REVIEW PACKAGE

LD405 pulse driver

TTL-gated constant-current driver for a 405 nm laser diode (BDR-209), the second, ionising step of two-photon REMPI during ablation loading of an ion trap

Board revision 1.0 · 30 October 2026 · built from commit `ed5fcb5` of `ParsaIQT/Fiber-solarisation`, folder `laser-pulse-driver/`

Pre-production: nothing has been built yet. Boards to be assembled by JLCPCB.



Output	high-side current source into the diode anode; cathode at ground (Thorlabs CG DE-9 pin-out)
Pulse current	0–600 mA (PULSE knob), 0–700 mA with solder jumper JP1; selected while TTL IN is high
Idle current	5–104 mA (IDLE knob) while TTL IN is low or open
Timing (simulated)	~0.7 µs delay, 4–6 µs 10–90 % rise, no overshoot, within 1 % after ≤ 12 µs
Protection	relay short + MOSFET crowbar when disarmed, soft start, over-current / compliance / 85 °C trips, two interlocks, latched fault, no automatic start
Hardware	160 × 100 mm 4-layer PCB, 138 parts (72 BOM lines), JLCPCB assembly; 9 V adapter; 3D-printed case
This package	this PDF and LD405_review_pack.zip (KiCad project, JLCPCB files, layer plots, check reports, simulation, enclosure, review renders; see Appendix B)

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1. What we would like reviewed

This driver fires a ~0.9 W, 405 nm laser diode for ~100 µs per ablation shot, gated by an ARTIQ TTL output. The diode has no published datasheet and is not cheap to replace, so the design puts the diode's safety first: the output is shorted whenever the driver is not armed, every fault latches, and the current source is linear and slow enough to stay stable with any reasonable cable.

Nothing has been built yet. Before the order goes to JLCPCB we would value a second opinion, above all on the points below. Section 10 lists what has and has not been verified so far.

Point 1 **Diode safety in every state.** Is there any sequence in which the diode could see a current spike, a reverse voltage or ESD? Consider power-up, power-down, the adapter unplugged or plugged in, ENABLE toggled, an interlock opened, the cable hot-plugged and the TTL lost. §4.5, §5

Point 2 **Arm/disarm order.** K1 is meant never to switch current. On disarm, the crowbar and the gate clamp act before K1 recloses. On arm, the loop starts 20 ms after K1 is energised. Is this robust, including relay bounce and the 74HC logic while the 5 V rail ramps? §5.2

Point 3 **Loop stability.** The current source was simulated with vendor models over FET transconductance ×0.5…×2 and 0.35–2 µH of cable. Are the margins convincing? Is anything important missing from the model, such as the real diode impedance, the op-amp output stage near the rail, or longer cables? §6

Point 4 **Limits for an unspecified diode.** The BDR-209 has no datasheet. Are these choices sensible?
- current ranges of 0–600 and 0–700 mA;
- a fixed 804 mA over-current trip that does not follow JP1;
- the compliance trip;
- no limit on pulse length (a stuck-high TTL gives CW). §2, §5

Point 5 **Layout.** The laser current path, the Kelvin sense at R40, the plane and return-current strategy, and Q2's heat sinking into board copper. The board was routed by a custom script and is DRC-clean. §7

Point 6 **Grounding.** Board ground is the diode cathode (DE-9 pin 3), the BNC shells and the DC-jack negative. Once ARTIQ and a scope are connected, it is tied to lab earth. Is there a ground-loop or common-mode concern, for example if the diode mount touches the optical table? §8.4

Point 7 **TTL input.** The input is high impedance: 10 kΩ to ground, 220 Ω in series with Schottky clamps, then a 74AHCT1G14 Schmitt trigger. It is driven from a Kasli DIO-BNC at 3.3 V or 5 V. Should there be a 50 Ω option? §4.3

Point 8 **Anything else.** Anything that would stop you ordering: part choices, footprints, JLCPCB assembly, the enclosure.

2. Requirements and specification

2.1 Requirements

- Second (ionising) step of a two-photon REMPI scheme during ablation loading: a BDR-209 405 nm Blu-ray diode run at roughly 0.9 W optical for ~100 µs per ablation shot, timed by an ARTIQ TTL output.
- “Plug and play”: fully assembled by a PCB house, in a box, mains powered (UK socket within 2 m of the diode).
- The diode must survive mistakes: hot-plugging, open or shorted cables, a lost TTL cable, power cycling and over-temperature.

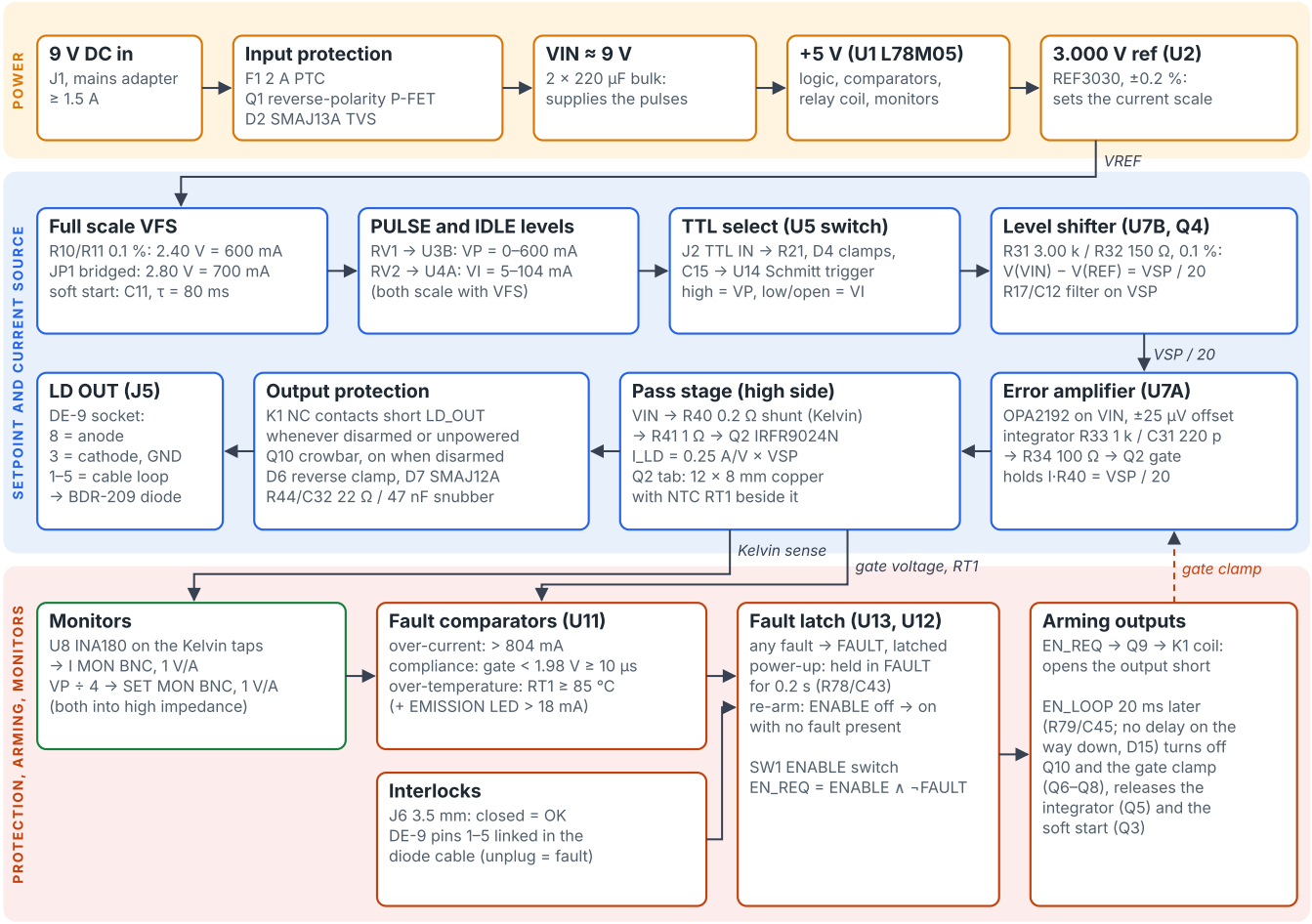
2.2 Design choices that follow

- **Constant current, cathode grounded, high-side source.** The diode module's case/cathode can sit at ground, and the DE-9 pin-out matches Thorlabs CG mounts (pin 8 anode, pin 3 cathode, pins 1–5 interlock loop).
- **Idle current between pulses** instead of switching the current fully off. The loop never leaves regulation, so pulses start within ~1 µs without the overshoot a saturated loop would produce. An idle current just above threshold doubles as a weak alignment beam.
- **Two independent setpoints** (PULSE, IDLE) selected by the TTL input through an analogue switch, so the TTL only has to switch, not shape the pulse.
- **Nothing digital in the current path**; the protection is hard-wired logic.

2.3 Specification

	value
Output	high-side current source, diode cathode at ground (Thorlabs “CG” DE-9 pin-out)
Pulse current	0–600 mA (PULSE knob); 0–700 mA with JP1 bridged; over-current trip at ~800 mA
Idle current	5–104 mA (IDLE knob) whenever TTL IN is low or unconnected
Timing (simulated, 0.5–2.5 m cable)	~0.7 µs delay, 4–6 µs 10–90 % rise, no overshoot, 1 % settled in ≤ 12 µs, ~7 µs fall
TTL input	BNC, 10 kΩ to ground, Schmitt trigger: high ≥ 2.0 V, low ≤ 0.5 V; 220 Ω + Schottky clamps to the rails; 3.3 V or 5 V TTL from ARTIQ (Kasli DIO-BNC / DIO-SMA)
Monitors	I MON (measured diode current) and SET MON (pulse setpoint), 1 V/A into high impedance (0.5 V/A into 50 Ω)
Compliance	≥ 7 V across the diode at 700 mA (the BDR-209 is reported to need ~5–6 V)
Protection	output shorted by a relay and a MOSFET crowbar whenever disarmed; soft start (τ = 80 ms); over-current, open-circuit/compliance and over-temperature (85 °C) trips; two interlocks (3.5 mm jack, DE-9 loop); latched fault that needs a deliberate re-arm; no start after power-up
Supply	9 V DC ≥ 1.5 A (5.5/2.1 mm, centre positive; Mean Well GST25A09-P1J suggested); reverse-polarity, over-voltage and PTC protection
Board, case	160 × 100 mm 4-layer PCB; 3D-printed case 181 × 108 × 32 mm (MJF PA12)
Diode	BDR-209 (DTR's Laser Shop, copper module). No datasheet; users report ~0.8–1 W at ~500–650 mA and ~5–6 V. The driver's limits are set from these reports.

3. Architecture



Block diagram. Reference designators match the schematic (Appendix C). Dashed: the gate clamp released by the arming logic; K1 and Q10 are driven by the same logic (see §5.2).

The pass element is a P-MOSFET (Q2) between the 9 V rail and the diode anode, with the 0.2 Ω shunt R40 and a 1 Ω degeneration resistor R41 in its source. An OPA2192 error amplifier, powered from VIN so its rail-to-rail inputs can sit at the shunt, compares the shunt voltage with a reference that a second OPA2192 section sets to VSP/20 below VIN. The setpoint VSP is either PULSE or IDLE, chosen by the TTL input through an analogue switch. Hard-wired logic latches every fault and controls three independent ways of keeping current out of the diode: the relay K1 (normally closed across the output), the crowbar Q10, and a clamp that holds Q2's gate at VIN.

4. Circuit description

Reference designators refer to the schematic (Appendix C). Values come from `tools/design.py`, which generates both the schematic and the PCB, so the two always agree.

4.1 Power (J1, F1, Q1, D1, D2, U1, U2)

9 V from a mains adapter → 2 A PTC (F1) → reverse-polarity P-FET Q1 (AO3401A; R1 holds its gate at 0 V and the Zener D1 limits V_{GS} to −10 V) → `VIN`. The SMAJ13A TVS D2 clamps `VIN`, and 2 × 220 µF plus ceramics buffer it and supply the pulse current. U1 (L78M05) makes `+5V` for the logic, comparators, monitors and relay. U2 (REF3030) makes the 3.000 V reference. The current-source op-amp (OPA2192) runs from `VIN`, so its rail-to-rail inputs see the shunt directly.

4.2 Setpoints (U2, R10/R11, JP1, RV1, RV2, U3, U4A, U5)

`VFS` = 3.000 V × 40.2 k / (10 k + 40.2 k) = 2.402 V, which corresponds to 600 mA (0.1 % resistors). Bridging JP1 puts R12 (4.02 k) in parallel with R10, giving `VFS` = 2.80 V, or 700 mA. `VFS` also charges C11 (10 µF) through R10 ‖ R11 = 8 k.

That is an 80 ms **soft start** whenever Q3 (driven by `DIS_LOOP`) releases it.

- PULSE: RV1 across 0…`VFS`, buffered by U3B → `VP` (0–600 mA).
- IDLE: R15 (100 k) – RV2 (20 k) – R16 (1 k) divider from `VFS`, buffered by U4A → `VI` = 19.8–417 mV ↔ 5–104 mA.
- U5 (SN74LVC1G3157) connects `VP` (TTL high) or `VI` (TTL low) to `VSP`, filtered by R17/C12 (1 k / 220 pF, 0.22 µs).

4.3 TTL input (J2, R21, R22, C15, D4, U14)

R22 (10 k) holds the input low when nothing is plugged in, which gives the idle current. R21 (220 Ω) and the BAT54S D4 clamp the input to the 5 V rails, and C15 removes glitches shorter than ~20 ns. U14 (74AHCT1G14 Schmitt trigger, thresholds 0.9–2.0 V rising and 0.5–1.6 V falling) drives the switch's select line. The input is not terminated; ARTIQ's DIO-BNC drives it directly.

4.4 Current source (U7, Q2, Q4, R30–R41, NT1, NT2)

```
VIN ── R40 0.2 Ω ─┬─── R41 1.0 Ω ── S  Q2 IRFR9024N  D ── LD_OUT ── DE-9 pin 8 → diode → pin 3 ── GND
     (Kelvin NT1) └ (Kelvin NT2)      G ← R34 100 Ω ← U7A (integrator R33 / C31)
```

- **Level shifter** (U7B + Q4) forces `VSP` across R31 (3.00 k), and the same current flows through R32 (150 Ω) from the Kelvin `VIN` tap, so `V(VIN) - V(REF) = VSP/20`.
- **Error amplifier** U7A (OPA2192, rail-to-rail, ±25 µV maximum offset) integrates `REF - SRC_SNS` with R33 = 1 k and C31 = 220 pF, and drives the P-FET gate through R34 = 100 Ω. In regulation `I · 0.2 Ω = VSP/20`, so **I_{LD} = VSP × 0.25 A/V**.
- **R41 (1 Ω) source degeneration** makes the pass stage's transconductance nearly independent of the MOSFET. The spread in FET g_m and the cable inductance made earlier topologies ring. R41 costs 0.7 V of headroom at 700 mA, which the 9 V supply has (compliance ≥ 7 V).
- **Kelvin connections**: net ties NT1/NT2 take the sense voltage from the inner edges of R40's pads, so the error amplifier and the INA180 see only the shunt.
- **Accuracy**: shunt ±1 %, reference ±0.2 % and dividers 0.1 % keep the setpoint scale within ±1.5 %; offset < 0.2 mA.

4.5 Output protection (K1, Q10, D6, D7, R44/C32)

- **K1** (Omron G6K-2F-Y): both normally-closed contacts short `LD_OUT` to ground unless the driver is armed. The diode is therefore shorted whenever the unit is off, unpowered or faulted, which makes hot-plugging and ESD on the cable harmless.
- **Q10** (AO3400A crowbar) shorts the output within microseconds of a disarm or fault, before the relay contacts move.
- D6 (SS34) clamps reverse voltage, D7 (SMAJ12A) clamps forward over-voltage, and R44/C32 (22 Ω / 47 nF) damp the cable.

4.6 Gate clamp and loop enable (Q5–Q8, Q3)

The pass FET is **off unless explicitly released**. Q6 pulls the gate to `VIN`, and R37/R38 bias it on from `VIN` alone. Only when `EN_LOOP` is high does Q8 turn on Q7, which turns Q6 off, so losing the 5 V rail turns the current off. While disabled, Q5 pulls the integrator input so its output sits at the “off” rail (anti-windup), and Q3 discharges the soft-start capacitor.

4.7 Monitors (U8, U10A, U4B)

- I MON: INA180A1 (gain 20) on the Kelvin taps gives 4 V/A, which a 0.1 % divider reduces to 1 V/A. A buffer then drives the BNC through 49.9 Ω: **1 V/A** into high impedance.
- SET MON: the PULSE setpoint `VP` ÷ 4 → buffer → 49.9 Ω → BNC: 1 V/A. It shows the pulse level before firing.

5. Protection and arming

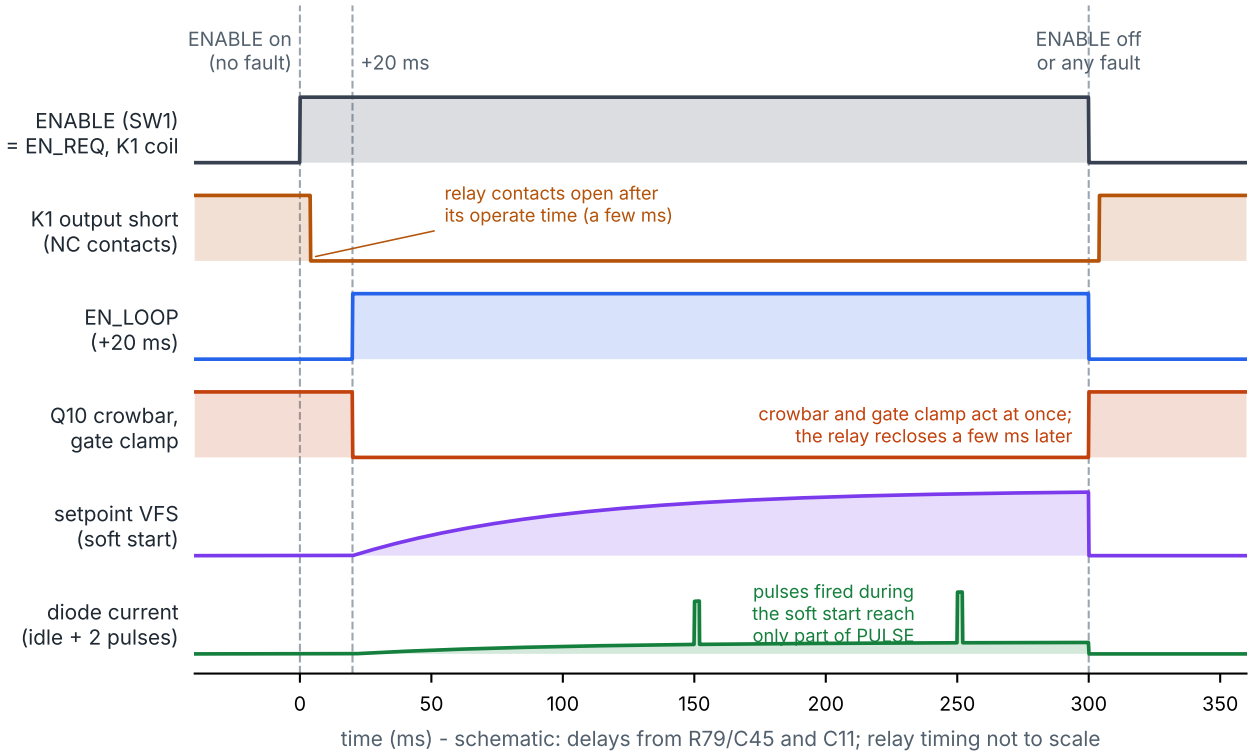
5.1 Fault sources

LM2901 open-collector comparators share the active-low fault line `S_N` (R73 pull-up, C44 filter). The interlocks join `S_N` through the BAT54A D10: the 3.5 mm jack (`REM_N`, closed = OK) and the DE-9 loop (`DB9_IL`, pins 1–5 linked in the cable). At power-up R78/C43 hold `S_N` low for ~0.2 s through D14.

source	trips when	threshold
U11A over-current	I MON (4 V/A) > `OC_REF` 3.21 V	804 mA
U11B compliance / open diode	gate < 1.98 V (loop saturated) for ≳ 10 µs (R52/R53/C33)	—
U11C over-temperature	NTC RT1 (10 k, B3950, beside Q2's tab) < 1.09 k	≈ 85 °C
Interlock, 3.5 mm jack J6	tip–sleeve open	—
Interlock, DE-9 loop	pins 1–5 not linked (cable unplugged)	—
Power-up hold	R78/C43 after +5 V appears	~0.2 s
(U11D emission, not a fault)	I MON > 74 mV: lights EMISSION, stretched to 0.1 s	≈ 18 mA

U13A/B form a NAND SR latch. Any fault sets `FAULT`, and it clears only when the ENABLE switch is turned off while no fault is present. `EN_REQ = ENABLE ∧ ¬FAULT` energises the relay at once. R79/C45 delay `EN_LOOP` by ~20 ms, until the relay contacts are open, and D15 removes the delay on the way down, so disarming is immediate. The loop then soft-starts with τ = 80 ms.

5.2 Arming and disarming



Arm/disarm sequence (schematic time scale). Arming opens K1's short first and releases the loop 20 ms later. Disarming, by the switch or by any fault, turns on the crowbar and the gate clamp at once, and K1 recloses a few ms later. The current never flows through a switching relay contact. Pulses fired in the first ~0.3 s after arming are reduced by the soft start.

5.3 Designed response to abnormal conditions

condition	designed response
Power applied	FAULT forced by the power-up hold (~0.2 s) and latched. K1 de-energised, so the output is shorted, and the gate clamp holds Q2 off. No automatic start.
ENABLE off → on, no fault	EN_REQ energises K1, which removes the short. 20 ms later EN_LOOP turns off the crowbar and the gate clamp and releases the integrator and soft start. The current ramps to the idle level (τ = 80 ms).
ENABLE switched off	EN_LOOP falls at once (D15). The crowbar and the gate clamp turn on, the integrator and soft start reset, and K1 releases and recloses its short a few ms later.
Interlock opened, or the DE-9 unplugged	S_N pulled low, FAULT latched, then as ENABLE off. Re-arming needs ENABLE off → on with the interlocks closed.
Open diode or broken cable while armed	The loop saturates and the gate falls below 1.98 V; after ~10 µs U11B trips → FAULT.
Over-current (> 804 mA)	U11A trips → FAULT. This is a backstop for a loop failure: normally the setpoint (≤ 700 mA) limits the current.
Q2 copper > ~85 °C	U11C (RT1) trips → FAULT.
TTL cable unplugged, or ARTIQ output high-Z	R22 pulls the input low → idle current.
TTL stuck high	Continuous current at the PULSE setting, with no time limit. Only Q2's 85 °C trip acts; the diode's own duty-cycle limit is not enforced (review point 4).
9 V adapter unplugged while armed	VIN and +5 V decay. K1 releases (output shorted), and the gate clamp, biased from VIN alone, holds Q2 off once EN_LOOP is lost. *To confirm on the bench: no current step while the rails collapse.*
+5 V lost, VIN still present	K1 releases (short); EN_LOOP low → Q6 clamps the gate from VIN.
Supply reversed / over-voltage	Q1 blocks reverse polarity; D2 (SMAJ13A) clamps VIN; F1 (2 A PTC) limits a sustained fault current.
Output shorted while armed	The source keeps regulating at the set current; Q2 then sees almost all of VIN (~5–6 W during a pulse). In pulsed use this is harmless; continuous operation ends with the 85 °C trip. A short is not flagged as a fault.
Q2 fails short	The current is no longer controlled: the over-current trip → FAULT → Q10 and K1 short the output, which protects the diode. The supply then feeds R40/R41/Q2/Q10 until F1 or the adapter's limit acts, and R41 may be overstressed.
Reverse voltage or ESD on the DE-9	Disarmed: shorted by K1. Armed: D6 (reverse) and D7 (12 V TVS) clamp; R44/C32 damp the cable.

> None of these sequences has been simulated at system level or tried on hardware. The bring-up plan (§11) exercises the main ones with a dummy load and a scope before the diode is connected.

6. Simulation

`sim/run_sims.py` runs the current source in ngspice with the vendor models (TI OPA2192 and INA180A1, Infineon IRFR9024N). The laser diode is modelled as a diode with 5–6 V at 0.6 A (N = 12, R_S = 1 Ω, C_J = 100 pF); it is not a measured BDR-209 model. The cable is 0.35 µH or 2 µH in *each* conductor (each with 0.15 Ω, and 100 Ω in parallel to model losses) and 150 pF across the diode, with the 22 Ω / 47 nF snubber at the driver. The testbench's component values were checked against `design.py` (R31, R32, R33, C31, R34, R35, R30, R40, R41, R17/C12, R44/C32). Corners: FET transconductance KP ×0.5…×2 (1.3, 2.6, 5.2), cable 0.35 µH and 2 µH per conductor (generous for 0.5 m and 2.5 m of twisted pair), supply 8.5–9 V, idle 5–100 mA, pulse 100–700 mA. The parallel 100 Ω adds damping that a real cable may not have; review point 3.

6.1 Large-signal 100 µs pulses (re-run 30 Oct 2026)

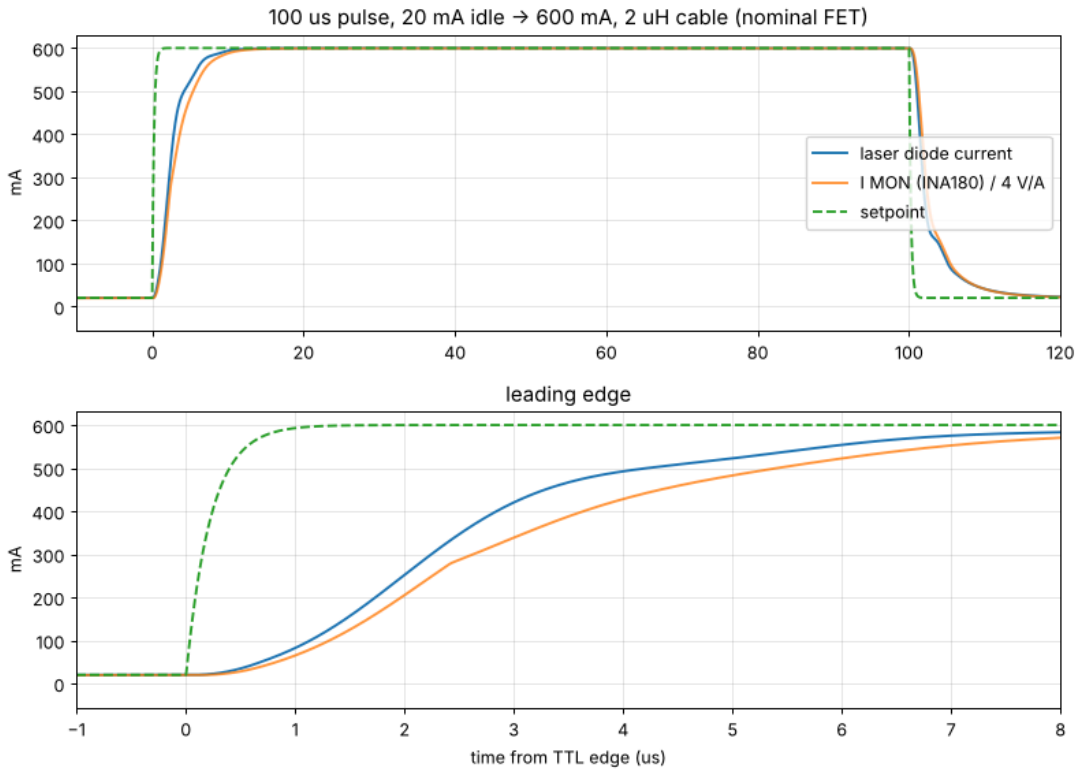
KP	L cable	VIN	idle → pulse	delay	10–90 % rise	overshoot	1 % settle	fall to 10 %
1.3	0.35 µH	9 V	19.9 → 600 mA	0.72 µs	5.68 µs	0.0 %	11.6 µs	8.89 µs
1.3	2 µH	9 V	19.9 → 600 mA	0.88 µs	5.43 µs	0.0 %	11.7 µs	8.85 µs
2.6	0.35 µH	9 V	19.9 → 600 mA	0.63 µs	4.85 µs	0.0 %	10.0 µs	7.45 µs
2.6	2 µH	9 V	19.9 → 600 mA	0.80 µs	4.75 µs	0.0 %	10.1 µs	7.41 µs
5.2	0.35 µH	9 V	19.9 → 600 mA	0.57 µs	4.26 µs	0.0 %	8.9 µs	6.43 µs
5.2	2 µH	9 V	19.9 → 600 mA	0.74 µs	4.42 µs	0.0 %	9.5 µs	6.41 µs
2.6	2 µH	9 V	4.9 → 600 mA	1.07 µs	4.66 µs	0.0 %	10.2 µs	6.55 µs
2.6	2 µH	9 V	19.9 → 700 mA	0.83 µs	4.65 µs	0.0 %	9.9 µs	6.94 µs
2.6	0.35 µH	8.5 V	19.9 → 700 mA	0.74 µs	4.59 µs	0.0 %	9.8 µs	7.08 µs
2.6	0.35 µH	9 V	19.9 → 100 mA	—	7.67 µs	0.0 %	14.3 µs	—
2.6	0.35 µH	9 V	99.9 → 600 mA	—	4.96 µs	0.0 %	9.5 µs	—

Delay is from the TTL edge to 10 % of the pulse current. "—": not defined when the idle current is above 10 % of the pulse (the last two rows).

6.2 Small-signal 5 % steps at 600 mA

KP	L cable	step	overshoot	2 % settle
1.3	0.35 µH	570 → 600 mA	0.0 %	9.2 µs
1.3	2 µH	570 → 600 mA	0.0 %	9.2 µs
2.6	0.35 µH	570 → 600 mA	0.0 %	8.1 µs
2.6	2 µH	570 → 600 mA	0.0 %	8.5 µs
5.2	0.35 µH	570 → 600 mA	0.0 %	7.3 µs
5.2	2 µH	570 → 600 mA	0.0 %	8.4 µs

A small step is not slew-limited, so its overshoot shows the phase margin: 0.0 % overshoot at every corner indicates a well-damped loop (more than ~70° phase margin).



Simulated 100 µs pulse, 20 mA idle → 600 mA, 2 µH cable, nominal FET: diode current, I MON and the setpoint.

The bandwidth (~80 kHz) is deliberately modest. It is fast enough for 100 µs pulses, with ~10 µs to full power, and slow enough to stay stable with any reasonable cable and FET. For much shorter pulses, reduce C31 and re-check stability with `run_sims.py`.

> **Not simulated:** the protection and arming logic (comparators, latch, relay, crowbar, power-up and power-down); thermal behaviour; the TTL front end (U14, U5) beyond an ideal setpoint edge; a measured diode model; cables longer than ~2.5 m.

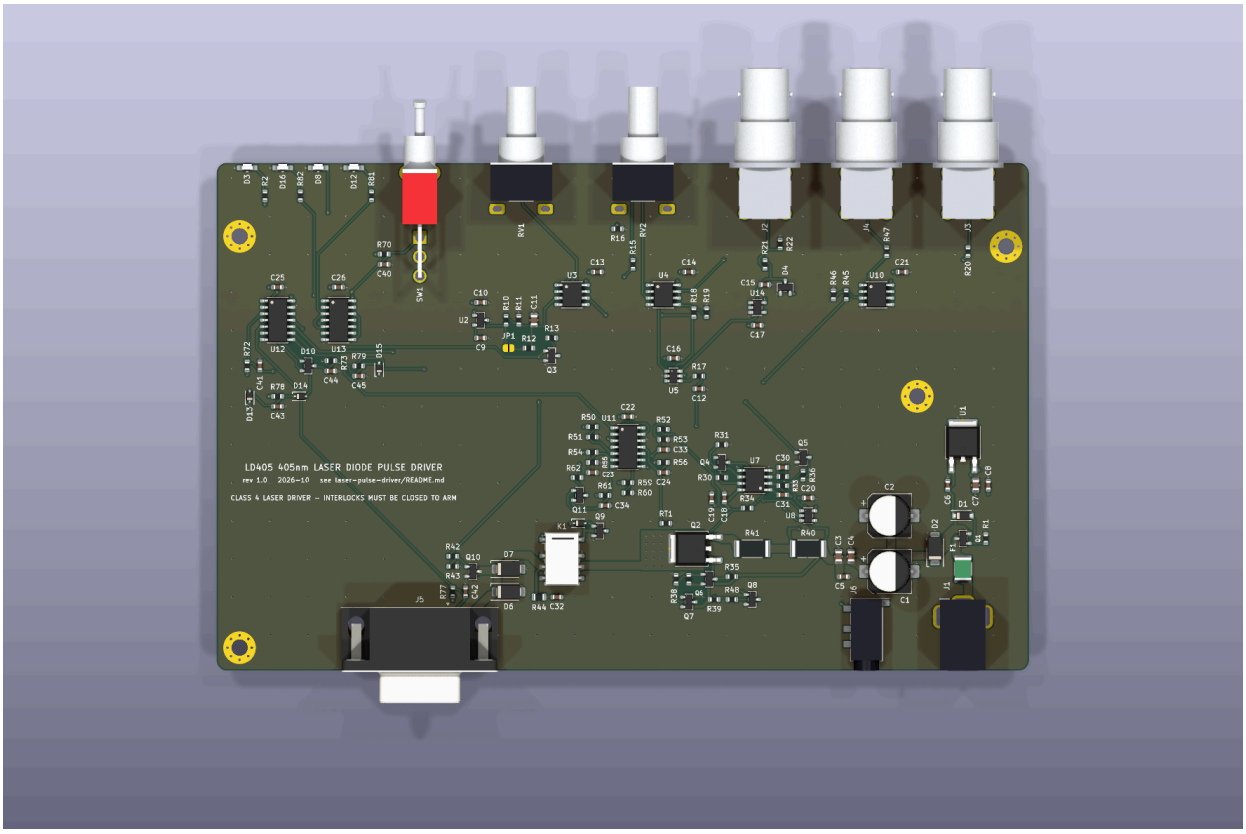
7. PCB

7.1 Stack-up and rules

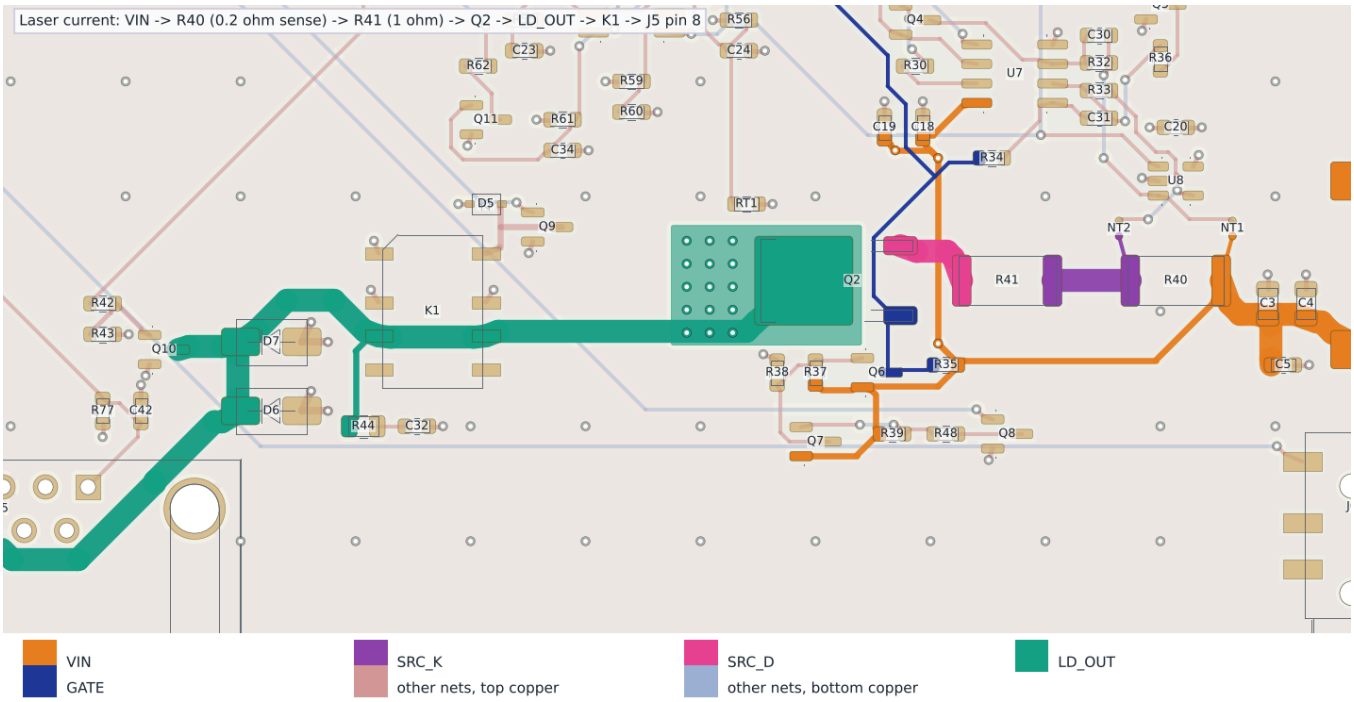
item	value
Size, layers	160 × 100 mm, 4 layers, 1.6 mm, JLC04161H-7628 stack-up; outer copper 1 oz, inner 0.5 oz
Layer use	F.Cu signals + GND pour · **In1 solid GND** · **In2 solid +5 V** · B.Cu signals + GND pour
Rules	0.15 mm minimum track and clearance; vias 0.3 mm drill / 0.6 mm pad, plugged; copper to edge 0.3 mm
Tracks	879 segments, 356 vias; widths 0.25–0.5 mm for signals, 1.5 mm for the laser current path
Planes	every GND/+5 V SMD pad has its own via to its plane; GND stitching vias every ~7.5 mm
Thermal	Q2's drain tab sits in a 12 × 8 mm LD_OUT copper area on both outer layers, joined by 15 thermal vias; the NTC RT1 is beside it

7.2 Layout features

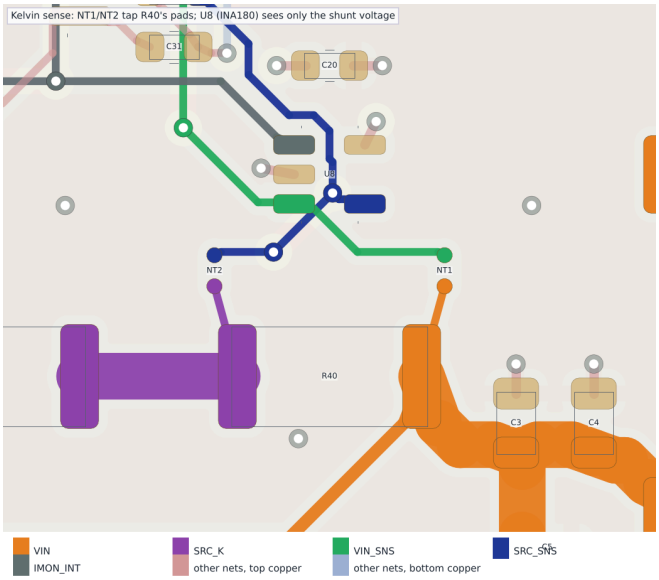
The laser current path runs from the input through the bulk capacitors, R40, R41, Q2 and the relay/clamps to the DE-9. It is 1.5 mm copper on F.Cu, necked only at the pads, directly above the GND plane that carries the return current. All panel connectors sit on the board edges. BNCs, pots, LEDs and the toggle are on the front edge, with their flanges or mounting faces on the edge line. The DE-9, DC jack and interlock jack are on the rear edge. M3 holes H1–H4 match the enclosure standoffs.



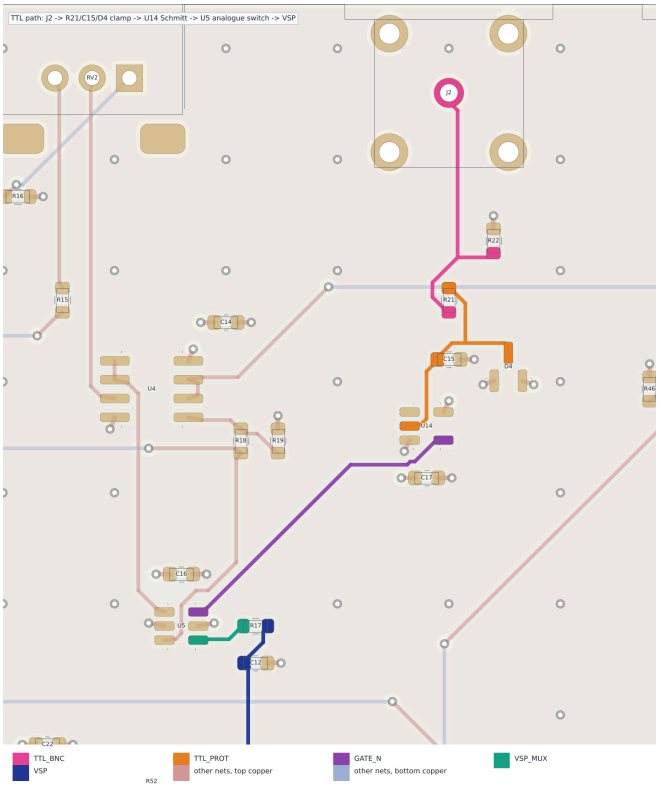
Top view (3D render; BNCs, pots, switch, jacks and side LEDs use simplified models).



Copper carrying the laser current, top and bottom layers.



R40 and its Kelvin sense pair to U8 and the error amplifier: net ties NT1/NT2 sit on R40's own pads, so the sense pair carries no load current.



TTL input path from J2 through R21/D4/C15 to U14 and U5.

7.3 Checks (run 30 Oct 2026, KiCad 9.0.9)

- ERC: 0 errors, 0 warnings.
- DRC with schematic parity: 0 violations, 0 unconnected pads, 0 footprint errors, no parity issues. The `lib_footprint_mismatch` check is set to *ignore*. The generator re-places reference designators and trims footprint silkscreen at the board edge, so no footprint is byte-identical to its library copy.

7.4 How the board was generated

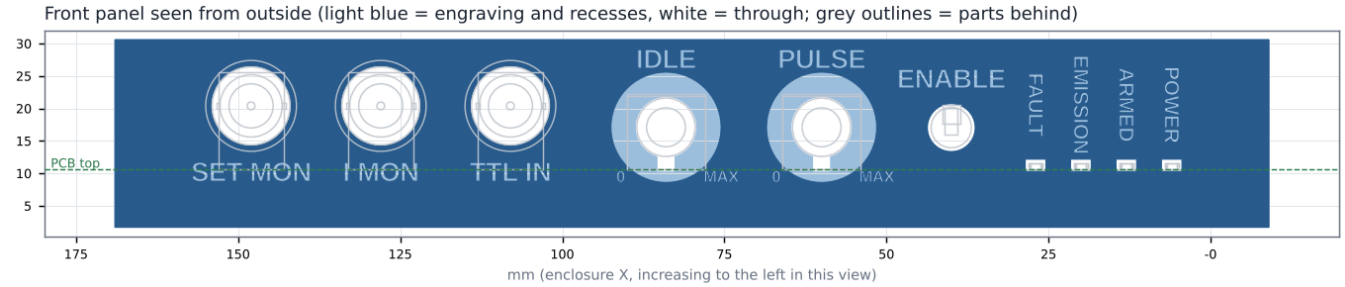
The schematic and the board are generated by scripts from one parts-and-nets file (`tools/design.py`) and a placement file. Routing was done by a purpose-written two-layer A* maze router on a 0.1 mm grid with clearance-inflated obstacle maps from the real pad shapes. It uses tapered power tracks, pre-routed Kelvin taps, plane fan-outs and rip-up-and-retry. Zones are then filled by KiCad. The result is checked only by KiCad's DRC and parity check and by the renders in this document. A human eye on the routing is therefore particularly welcome (review point 5).

8. Connectors, cable, enclosure, grounding

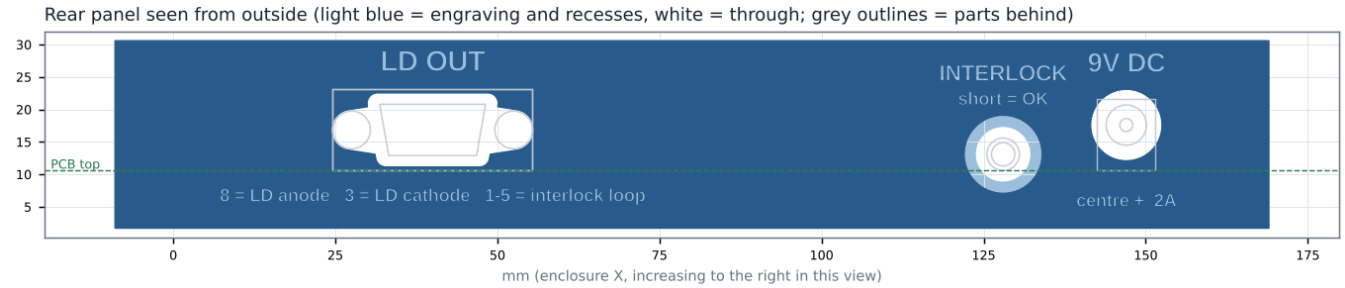
8.1 Panels

front	function
TTL IN (BNC)	high = pulse current, low/unconnected = idle current
I MON (BNC)	measured diode current, 1 V/A (high-Z scope input)
SET MON (BNC)	pulse setpoint, 1 V/A; shows what a pulse will deliver before firing
PULSE, IDLE (knobs)	pulse current 0 → full scale (600 or 700 mA); idle current 5 → 104 mA
ENABLE (toggle)	arm/disarm, and reset after a fault (off → on)
LEDs	POWER (green); ARMED (yellow: relay open, source enabled); EMISSION (blue: > ~18 mA, stretched so 100 µs pulses are visible); FAULT (red, latched)

rear	function
LD OUT (DE-9 female)	8 = anode, 3 = cathode/ground, 1–5 = interlock loop (Thorlabs CG compatible)
INTERLOCK (3.5 mm)	must be closed (tip–sleeve shorted) to arm
9V DC	9 V, ≥ 1.5 A, centre positive

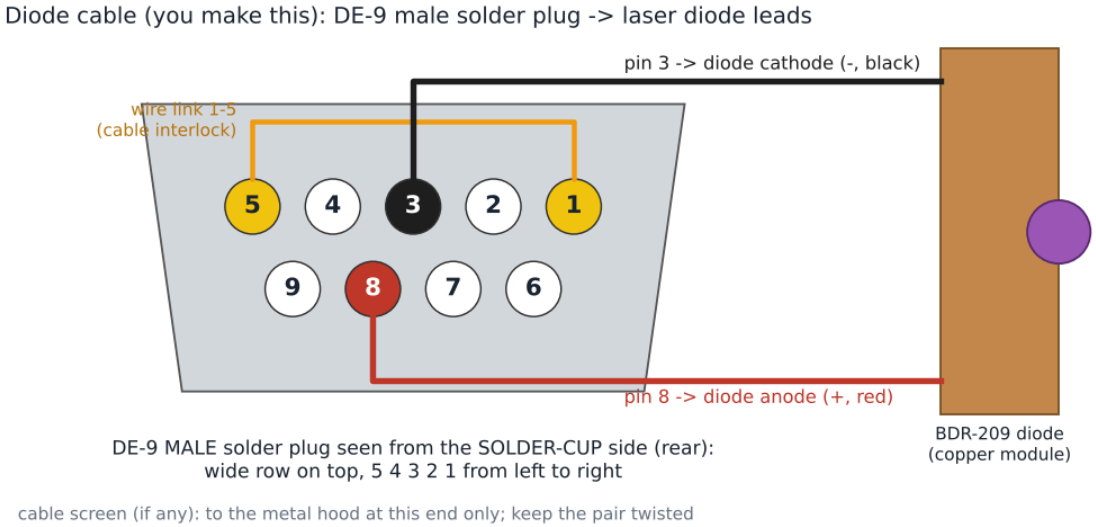


Front panel from outside (white: through; light blue: engraving and recesses; grey: the parts behind).



Rear panel from outside.

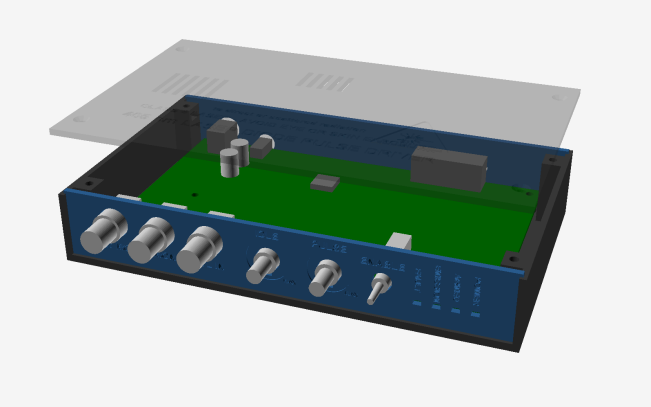
8.2 Diode cable



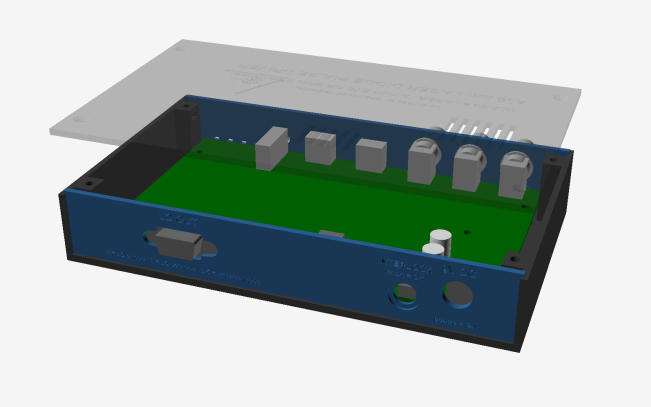
The diode cable, made by the user: DE-9 male, pin 8 → anode, pin 3 → cathode, wire link 1–5 inside the hood (unplugging trips the driver); screen, if any, to pin 3 at the driver end only.

8.3 Enclosure

The OpenSCAD model is generated from the board placement, with connector axis heights from the datasheets. The front and rear panels are 2.5 mm plates held in grooves by the base and lid, so they are fitted to the board's connectors before the board goes into the base. To be printed in MJF PA12 at JLC3DP. JLC3DP's thin-wall heatmap flagged the lid posts, so all four parts were reworked and checked for walls under 1.2 mm; only the ridges between engraved letters remain. Hardware: 4 × M3×16 countersunk with nuts (PCB), 4 × M3×10 countersunk with nuts (lid).



Assembly preview, front.



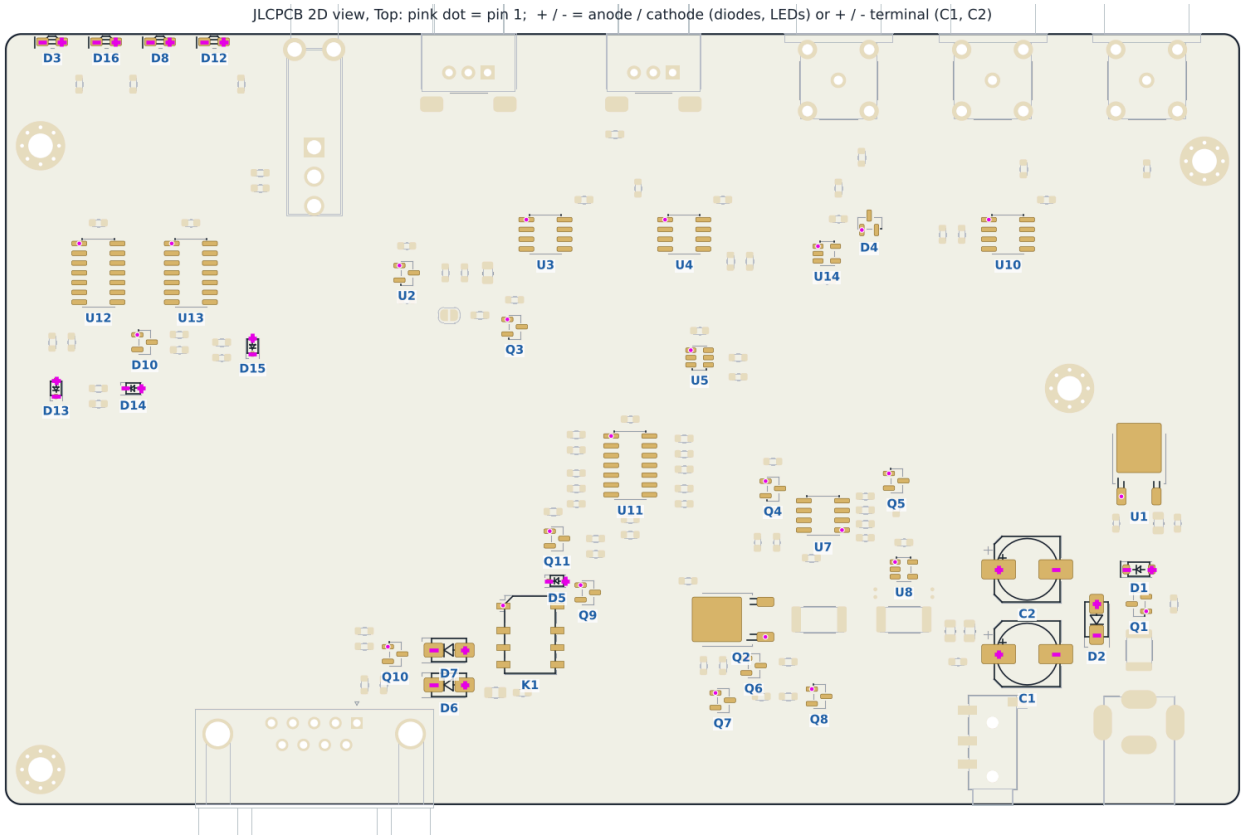
Assembly preview, rear.

8.4 Grounding

Board GND is common to the diode cathode (DE-9 pins 3 and 5 and the shell), the shells of the three BNCs, the 3.5 mm jack sleeve and the DC-jack negative. The 9 V adapter output is SELV-isolated from the mains; whether its negative is tied to mains earth depends on the adapter. In use, the BNC shells connect the board to ARTIQ and to the scope, and so to lab earth. The instructions say the diode module's case must not touch anything except through the cable, because the cathode is ground. This is review point 6.

9. Manufacturing

item	choice
PCB	JLCPCB: 4 layers, 1.6 mm, 1 oz outer / 0.5 oz inner, lead-free HASL (or ENIG), vias plugged, min via 0.3 mm
Assembly	JLCPCB Standard PCBA, top side only. The through-hole connectors, pots and toggle are hand-soldered by JLCPCB. JP1 is left open (600 mA range).
BOM	72 lines, 138 parts (138 placements), all with LCSC part numbers; BOM and CPL uploaded to JLCPCB and checked in its placement preview on 29 Oct 2026
Placement	CPL rotations include the usual KiCad → JLC offsets (SOT-23 +180°, SOIC +270°), offsets for the panel parts (BNCs +90°; pots, toggle and DE-9 +270°) and per-part corrections found in JLCPCB's placement preview (C1/C2 0° rather than the usual +180° for electrolytics, K1 +270°, Q2 +90°, D10 +270°)
Enclosure	JLC3DP, MJF PA12 (black): base, lid, front and rear panels (STL in print orientation)
Off-board	Mean Well GST25A09-P1J 9 V adapter; DE-9 male plug + hood; 3.5 mm mono plug; knobs for 6 mm D shafts; M3 screws and nuts; rubber feet



Where JLCPCB's placement-preview markers must land: a dot on pin 1, and + / – on anodes, cathodes and the electrolytics' terminals. The owner compared JLCPCB's preview with this map; the rotation corrections in the table above came out of that check.

10. Verification status

check	how	result
Electrical rules	KiCad ERC (30 Oct 2026)	0 errors, 0 warnings
Layout rules, connectivity	KiCad DRC with schematic parity (30 Oct 2026)	0 violations, 0 unconnected, parity clean
Current source	ngspice with vendor models, 11 large-signal + 6 small-signal corners (§6)	no overshoot, settles in ≤ 12 µs at all corners; re-run for this package with identical results
Pinouts of active parts	renders of every footprint with pad numbers and nets, compared with the manufacturers' drawings by the project owner (review items A1–A7)	A1–A3, A5, A6 confirmed. A4 (REF3030 drawing labels) and A7 (side LEDs: the original part's datasheet contradicted itself, so the part was replaced) were flagged and fixed, not yet re-confirmed.
Panel-part footprints	hole sizes and pin positions vs datasheets (B1–B6)	confirmed
Polarity / pin 1 of 40 parts	polarity map and CPL rotations (A2), then JLCPCB's placement preview	confirmed; rotation corrections for C1/C2, K1, Q2 and D10 made after the preview (§9)
Enclosure fit	cross-sections through every panel hole, panel faces, lid clearance (C1–C5)	C1, C3, C5 confirmed. C2 (toggle hole, redrawn and widened) and C4 (DC jack hole 0.5 mm low) were flagged and fixed, not yet re-confirmed.
Schematic, current path, Kelvin sense, TTL input, diode cable	owner's review of renders (D1–D4, E1)	confirmed
3D-print printability	JLC3DP thin-wall heatmap + own voxel thickness checks	no wall under 1.2 mm except the ridges between engraved letters

Not verified: no hardware has been built; there is no measured diode data or datasheet; the protection and arming logic, the power-up and power-down transients and the thermal behaviour are reasoned, not simulated or measured; EMC has not been assessed; and the routing is checked only by DRC and the renders above. It is also inferred, not confirmed, that “lever up” on the ENABLE toggle (Dailywell M7) is the 2–1 (ENABLE) position. The bench plan below covers most of this before the diode is connected.

11. Bring-up and test plan

1. **Without the diode.** Plug in the DE-9 cable with a **dummy load** instead of the diode (5 × 1N5408 in series plus 1 Ω 2 W, ≈ 5–6 V at 0.6 A; pulsed use only) and the interlock plug. Turn both knobs fully anticlockwise.
2. Power up: POWER on, FAULT on. ENABLE off → on: ARMED on, FAULT off.
3. Scope on I MON (1 V/A). Turn IDLE: the trace moves between ~5 and ~104 mV.
4. Drive TTL IN with 100 µs pulses at ≤ 10 Hz. Turn PULSE up: I MON shows the pulses, and SET MON shows the level a pulse will reach. Check that the pulse at full scale (≈ 600 mV on I MON) is flat with no ringing.
5. Optional 700 mA range: bridge JP1 (near U2).
6. Disconnect one lead of the dummy load at the far end with the DE-9 still plugged in: FAULT must latch (open-circuit detection). Unplugging the DE-9 must also latch FAULT (cable interlock).
7. **Suggested additions for this review:** with a second scope channel directly across the dummy load, capture LD_OUT during arm, disarm, an interlock opening, the 9 V adapter being unplugged while armed and while pulsing, and plugging the adapter back in. No step or spike should appear beyond the set current. Check that I MON agrees with a current probe or shunt, and that Q2's copper stays cool at the intended duty cycle.
8. Set both knobs to minimum, **disarm**, connect the diode, then arm. Calibrate optical power with a thermal power meter at a known duty cycle ($\bar{P} \approx P_{peak} \cdot D$) or with a fast photodiode. Increase PULSE slowly while reading I MON, and do not exceed the diode's current rating.

12. Known limitations

- Pulse length is unlimited (CW works), but at 600–700 mA continuous the pass transistor dissipates ~2 W and will eventually trip the 85 °C sensor; use a low duty cycle as intended.
- The minimum idle current is ~5 mA (the diode glows faintly). For a darker idle, replace R16 (1 k) with 0 Ω (IDLE then spans 0–100 mA).
- The 3.5 mm interlock input expects a closed contact; the DE-9 interlock requires pins 1 and 5 linked in the diode cable.
- The relay is rated 1 A. Do not raise the current range above ~800 mA without re-checking K1, R40/R41 dissipation and Q2's SOA.
- The ENABLE toggle is held by the board only (its bushing passes through the panel without a nut).

Appendix A. Bill of materials

72 lines, 138 parts; JLCPCB/LCSC part numbers as uploaded. File: `manufacturing/ld405-driver_bom_full.csv`.

qty	value	designators	manufacturer part	LCSC
2	220uF 25V	C1 C2	RVT1E221M0810	C72497
4	10uF	C3 C4 C7 C11	CL21A106KAYNNNE	C15850
17	100nF	C5 C8 C13 C14 C16 C17 C18 C20 C21 C22 C23 C24 C25 C26 C34 C40 C41	CC0603KRX7R9BB104	C14663
4	1uF	C6 C9 C10 C19	CL10A105KB8NNNC	C15849
2	220pF C0G	C12 C31	GCM1885C2A221JA16D	C388907
3	100pF C0G	C15 C30 C33	CL10C101JB8NNNC	C14858
1	47nF	C32	CL10B473KB8NNNC	C1622
1	10nF	C42	0603B103K500NT	C57112
1	2.2uF	C43	CL10A225KO8NNNC	C23630
1	1nF	C44	CL10B102KB8NNNC	C1588
1	220nF	C45	CL10B224KA8NNNC	C21120
1	BZT52C10	D1	BZT52C10	C19077408
1	SMAJ13A	D2	SMAJ13A	C19077533
1	POWER green	D3	LTST-S220TGKT	C160470
1	BAT54S	D4	BAT54S	C7420333
4	1N4148WS	D5 D13 D14 D15	1N4148WS	C2128
1	SS34	D6	SS34	C8678
1	SMAJ12A	D7	SMAJ12A	C19077531
1	EMISSION blue	D8	LTST-S220TBKT	C160471
1	BAT54A	D10	BAT54A	C7502690
1	FAULT red	D12	LTST-S220KRKT	C284935
1	ARMED yellow	D16	LTST-S220KSKT	C160468
1	2A PTC	F1	BSMD1812-200-30V	C960026
1	9V DC IN	J1	DC-005-5A-2.0	C381116
1	TTL IN	J2	DOSIN-801-0080	C521219
1	SET MON	J3	DOSIN-801-0080	C521219
1	I MON	J4	DOSIN-801-0080	C521219
1	LD OUT	J5	DS1037-09FNAKT74-0CC	C75749
1	INTERLOCK	J6	PJ-320D	C431535
1	G6K-2F-Y 5V	K1	G6K-2F-Y-TR DC5	C47190
4	AO3401A	Q1 Q6 Q7 Q11	AO3401A	C15127
1	IRFR9024N	Q2	IRFR9024NTRPBF	C2585
5	2N7002	Q3 Q4 Q5 Q8 Q9	2N7002	C8545
1	AO3400A	Q10	AO3400A	C20917
11	100k	R1 R15 R36 R37 R39 R43 R53 R54 R59 R78 R79		C25803
1	2.2k	R2		C4190
1	10k 0.1%	R10	RT0603BRD0710KL	C95204
1	40.2k 0.1%	R11	TD03G4022BT	C667101
1	4.02k	R12		C23040
4	100	R13 R30 R34 R42		C22775

qty	value	designators	manufacturer part	LCSC
6	1k	R16 R17 R33 R62 R81 R82		C21190
3	3k 0.1%	R18 R31 R45	RT0603BRD073KL	C136963
2	1k 0.1%	R19 R46	RT0603BRD071KL	C110776
2	49.9	R20 R47		C23185
1	220	R21		C22962
6	10k	R22 R35 R50 R56 R70 R73		C25804
1	150 0.1%	R32	AR03BTCX1500	C319859
2	47k	R38 R48		C25819
1	0.2	R40	RLP25FEER200	C459674
1	1.0	R41	25121WF100KT4E	C11254
1	22	R44		C17561
1	18k	R51		C25810
1	300k	R52		C23024
1	11k	R55		C25950
1	1.5k	R60		C22843
1	1M	R61		C22935
1	4.7k	R72		C23162
1	3.3k	R77		C22978
1	10k B3950	RT1	SDNT1608X103F3950FTF	C279936
1	PULSE 20k	RV1	RK09L112003P	C470444
1	IDLE 20k	RV2	RK09L112003P	C470444
1	ENABLE	SW1	1MS1T2B4M7QES	C496157
1	L78M05ABDT	U1	L78M05ABDT-TR	C58069
1	REF3030	U2	REF3030AIDBZR	C38423
3	TLV9062	U3 U4 U10	TLV9062IDR	C398355
1	SN74LVC1G3157	U5	SN74LVC1G3157DBVR	C10426
1	OPA2192	U7	OPA2192IDR	C110074
1	INA180A1	U8	INA180A1IDBVR	C122228
1	LM2901	U11	LM2901DR	C7932
1	74HC14	U12	74HC14D	C5605
1	74HC132	U13	SN74HC132DR	C350560
1	74AHCT1G14	U14	SN74AHCT1G14DBVR	C55250

Appendix B. Contents of LD405_review_pack.zip

folder	contents
`00_README.txt`	what this package is and where to start
`LD405_design_review.pdf`	the review document: start here
`schematic/`	ld405-driver_schematic.pdf (A0, vector)
`pcb/`	layer plots (F.Cu, In1.Cu, In2.Cu, B.Cu, silkscreen, fab), assembly drawing, 3D renders
`manufacturing/`	the files uploaded to JLCPCB: Gerber/drill zip, BOM, CPL (with rotation offsets), full BOM
`kicad/`	KiCad 9 project (schematic, PCB, project file, custom footprints and 3D models)
`checks/`	KiCad ERC and DRC reports from this build

folder	contents
`simulation/`	ngspice testbench and corner script, results of this build, simulated pulse plot; vendor models are fetched by get_models.sh
`enclosure/`	OpenSCAD source, STL files (print orientation), previews
`review_renders/`	the review renders: footprints vs datasheets, polarity and JLC marker maps, copper close-ups, panel cross-sections, diode cable (index in README.md)
`docs/`	design notes, ordering and assembly guide (README), ARTIQ example
`source/`	the generator scripts (design.py holds every part, value, LCSC number and connection)

Appendix C. Schematic and assembly drawing

The next page is the complete schematic as one A0 sheet (vector: zoom in). It is followed by the assembly drawing (top fab layer). Both are also in the zip as separate files.

The schematic is generated from `design.py`. It is drawn as functional blocks in which parts connect through net labels rather than wires, so the block diagram (§3) and the net names used in §4–§5 are the map to it. KiCad's net highlighting in the project (zip: `kicad/`) or `source/tools/design.py`, which lists every pin's net, may be quicker to follow.

1 POWER INPUT, 5 V AND 3.000 V REFERENCE

9 V / ≈+1.5 A desktop adapter (5.5/2.1 mm, centre +).
Q1 = reverse-polarity protection, F1 = resettable fuse, D2 = TVS.

J1 DC IN
F1 2A PTC
Q1 SI2302
D2 SMAJ5.0A
U1 TBH05A80T
C1 220uF 25V
C2 220uF 25V
C3 10uF
C4 10uF
C5 100nF
C6 1uF
C7 10uF
C8 100nF
C9 1uF
C10 1uF
U2 REF3030
PWR_FLAG
PWR_FLAG

2 SETPOINT, SOFT START AND TTL GATING (scale 4 V/A)

VFS = 2.40 V = 600 mA full scale (2P1 closed); 2.80 V = 700 mA).
C11 = 910 (91) gives an 80 ms soft start; Q3 discharges it when disabled.
U5 selects PULSE (TTL high) or IDLE (TTL low, 5.10k mA).

R10 10k 0.1%
R11 40.2k 0.1%
JP1 LMAX 700mA
R12 4.02k
C11 10uF
U3 TLV9062
U3B TLV9062
R13 100
C12 220pF C0G
R14 20k
R15 100k
C13 100nF
U4A TLV9062
R16 1k
R17 1k
C14 100nF
C15 100nF C0G
R18 100k
C16 100nF
U5 74LVC1G157
Q3 2N7002
C17 100nF
C18 100nF
C19 1uF
R30 330
C31 220pF C0G
R37 330k
R38 47k
Q2 IRF9022V4
R42 8.2
C20 100nF
D3 SS 83950

3 TTL INPUT (BNC, 10 k to GND, 3.3/5 V logic)

Schmitt trigger with TTL thresholds; unconnected input = IDLE.

J2 TTL IN
D4 BAT54S
U13 74AHCT1G14
R22 10k
R21 220
C15 100pF C0G
C17 100nF

4 HIGH-SIDE CURRENT SOURCE (laser cathode at GND)

LLD = VSP × (R32/R31) // Rhd = VSP / ≈ V/A.
U7A error amp senses Rhd directly (Kelvin via NTL/NCD); Rh1 degenerates Q2.
Q6 clamps Q2 off unless EN_LOOP is high (fail-safe; QT/QR release 1Ω.
Q5 forces U7A high while disabled so the integrator cannot wind up.

U7A OPA2192
U7B OPA2192
R31 3k 0.1%
R32 150 0.1%
C18 100nF
C30 100pF C0G
R35 10k
R36 100k
Q5 2N7002
Q6 2N7002
R39 100k
R48 47k
R49 1.0
R50 47k
R51 1.0
Q4 NDS9401A
Q7 NDS9401A
R71 RefTie
R72 RefTie
C19 1uF
R33 1k
C31 220pF C0G
R37 330k
R38 47k
Q8 2N7002
R42 8.2
C20 100nF
U8 INA180A1
D3 SS 83950
Q2 IRF9022V4
R34 330
R38 47k
R42 8.2
D3 SS 83950
Q4 NDS9401A
Q7 NDS9401A
R71 RefTie
R72 RefTie
R49 1.0
R51 1.0

5 OUTPUT, SHORTING RELAY AND CROWBAR

K1 NC contacts short the laser whenever the driver is not armed or unpowered.
Q10 crowbar shunts I_OUT within 1 µs of a fault; D6/D7 reverse and surge clamp.
D99 follows Thorlabs LD-driver 'CC' pinout: 8 = anode, 3 = cathode/GND, 1–5 interlocks.

K1 660 660 5 3 4 2 1
K1 5VDC-2F-Y-5V
Q9 2N7002
D6 SS34
D5 1N5148WS
Q10 NDS9410A
R43 100k
R44 22
R42 100
C32 47nF
D99 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
D99 DB15-HD-M
D7 SMAJ22A

6 CURRENT AND SETPOINT MONITORS (1 V/A into high-Z)

I_MON from INA180 (≈ V/A) // ≈ SET MON shows the PULSE setpoint // ≈.

U6A TLV9062
U10B TLV9062
U10A TLV9062
R8 3k 0.1%
R45 3k 0.1%
C21 100nF
R9 14 0.1%
R46 14 0.1%
R47 49.9
R20 48.9
J7 SET MON
J8 I_MON

7 PROTECTION COMPARATORS

A over-current = 800 mA. B: compliance / open laser (gate = 2 V for > 10 us).
C: Q2 heatsink > 85 C (NTC RT1). D: enclosure > 19 mA (gets the EMISSION LED for ~0.15 s.

U11A LMV901
U11B LMV901
U11C LMV901
U11D LMV901
U11E LMV901
R61 18k
R54 100k
R55 11k
R58 100k
R59 100k
R60 1.5k
R61 1k
C33 100pF C0G
C34 100nF
R56 10k
C35 100nF
C23 100nF
C24 100nF
R62 1k
D11 SML-D12V8W
R50 330
R52 300k
R53 100k

8 INTERLOCKS, FAULT LATCH AND ENABLE SEQUENCING

Any fault, open interlock or power-up sets the FAULT latch; reset by switching ENABLE off.
EN_REQ arms relay K1 at once; EN_LOOP releases the current source ~17 ms later.

SW1 ENABLE
R70 10k
C42 10nF
C40 100nF
J6 INTERLOCK
R13 4.7k
C41 100nF
R77 3.3k
U12A 74HC14
U12B 74HC14
U12C 74HC14
U12D 74HC14
U12E 74HC14
U12F 74HC14
D10 BAT54A
R73 10k
C44 1nF
R78 100k
C43 2.2nF
C25 100nF
D14 1N5148WS
U13A 74HC132
U13B 74HC132
U13C 74HC132
U13D 74HC132
D13 1N5148WS
R79 100k
C45 220nF
D15 1N5148WS

9 FRONT-PANEL INDICATORS

D3 POWER green
D16 ARMED yellow
R7 2.2k
R82 1k
D12 FAULT red
D8 EMISSION blue
FB1 1k

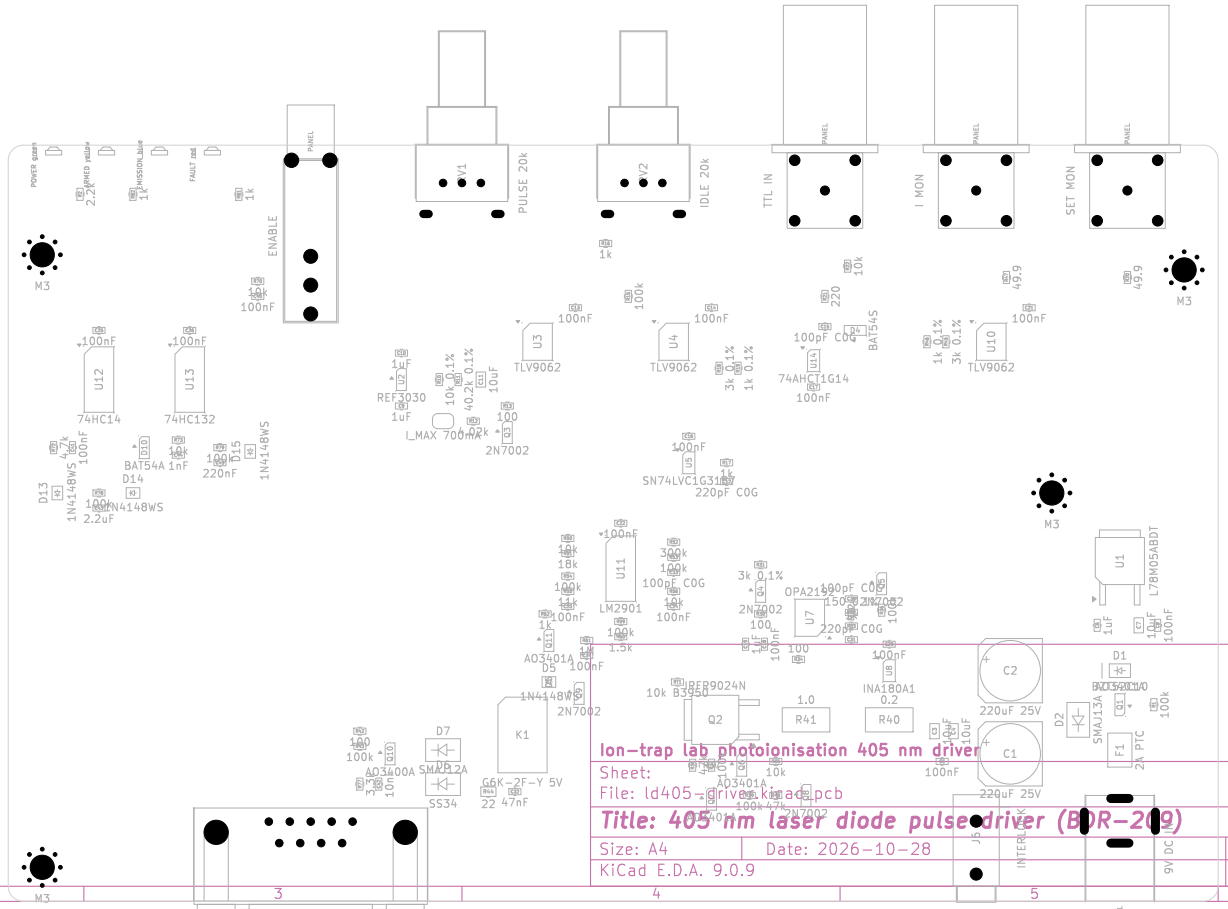
CLASS 4 LASER — read README safety section before use
Laser: DTR BDR-209 405 nm (500 mA nominal, 600 mA max), CATHODE AT GND
Generated from tools/design.py by tools/gen_sch.py
ion-trap lab photoionisation 405 nm driver
Sheet: /
File: ld405-driver.kicad_sch
Title: 405 nm laser diode pulse driver (BDR-209)
Size: A1 | Date: 2026-10-28 | **Rev: 1.0**
KiCad 10.0.1 | ID: 1/1

10 IC POWER PINS

U1C TLV9062
U4C TLV9062
U6B SN74LVC1G3157
U7C OPA2192
U10C TLV9062
U11E LMV901
U13E 74HC14
U13C 74HC132

11 MOUNTING HOLES

H1 M3
H2 M3
H3 M3
H4 M3



Ion-trap lab photoionisation 405 nm driver

Sheet:

File: ld405_driver_knacq.pcb

Title: 405 nm laser diode pulse driver (BDR-219)

Size: A4	Date: 2026-10-28	**Rev: 1.0**
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